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[23EE32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ELECTRONIC CIRCUITS AND APPLICATIONS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define forward static and dynamic resistances of diode.	L1	CO1	**02**
	b.	Define rectifier. List different types of rectifiers.	L1	CO2	**02**
	c.	List the differences NPN and PNP transistors.	L1	CO3	**02**
	d.	Mention advantages and Disadvantages of Power MOSFET.	L1	CO5	**02**
	e.	Mention two essential conditions for a circuit to maintain oscillations.	L1	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the diode approximations with neat diagram and explanation.	L2	CO1	**05**
		OR			
	b.	Explain the effect of temperature on diode characteristics.	L2	CO1	**05**
3.	**a.**	Explain the VI characteristics of Zener diode.	L2	CO2	**05**
		OR			
	b.	Compare Half Wave Rectifier and Full Wave Rectifier.	L3	CO2	**05**
4.	**a.**	Explain the NPN transistor structure and working operation.	L2	CO3	**05**
		OR			
	b.	Explain BJT common-base configuration with circuit diagram and sketch different common-base characteristics.	L2	CO4	**05**
5.	**a.**	Compare BJT and FET.	L3	CO4	**05**
		OR			
	b.	Analyse the gate bias circuit for the JFET.	L3	CO4	**05**
6.	**a.**	In a phase shift oscillator circuit, the values of R1=R2=R3=1MΩ and C1=C2=C3=68PF. At What frequency does the circuit oscillate?	L2	CO6	**05**
		OR			
	b.	In an RC phase shift oscillator R=1Ω, RC=1Ω and C=0.1μF. Calculate the frequency of oscillations.	L2	CO6	**05**

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7.	a.	Draw the circuit diagram of a positive clipper circuit and explain its working with necessary input and output waveform.	L2	CO1	05
		OR			
	b.	Discuss the class A power amplifier with necessary diagrams.	L2	CO5	05
		PART-C	**6 X 10=60**		
8.	a.	Draw the V-I characteristics of PN junction Diode and explain the diode parameters.	L2	CO1	10
		OR			
	b.	Explain the concept of DC load line with neat diagram and necessary equations.	L2	CO1	10
9.	a.	Sketch the circuit of a Half-Wave Rectifier and explain its operation. Derive Vrms.	L2	CO2	10
		OR			
	b.	Explain the working of Bridge rectifier with neat waveform.	L2	CO2	10
10.	a.	Sketch typical BJT common-emitter input and output characteristics. Explain the shape of the characteristics.	L2	CO3	10
		OR			
	b.	Explain the following terms and their significance. i) DC load line ii) Bias Point	L2	CO3	10
11.	a.	Draw a cross section diagram for an enhancement-mode MOSFET and explain its operation.	L2	CO4	10
		OR			
	b.	Draw a circuit diagram for obtaining the drain and transfer characteristic for n-channel JFET and explain.	L2	CO4	10
12.	a.	Discuss the working of class B power amplifiers with necessary diagrams.	L3	CO5	10
		OR			
	b.	Explain the basic operation of Buck converter and derive the expression for duty cycle, D.	L3	CO5	10
13.	a.	Illustrate the circuit of a RC Phase shift oscillator and discuss its working. Give its merits and drawbacks.	L3	CO6	10
		OR			
	b.	Derive the frequency of oscillations in Wein bridge oscillator by clearly explaining the operation of the same.	L3	CO6	10